Poisson Elastic Net: Nested CV

Hyperparameter Grid $7 \times 7 = 49$ combinations

$$\text{penalty} \in \{1.00e\text{-}10,\ 4.64e\text{-}09,\ 2.15e\text{-}07,\ 1.00e\text{-}05,\ 4.60e\text{-}04,\ 2.15e\text{-}02,\ 1.00\} \times \text{mixture} \in \{0.000,\ 0.167,\ 0.333,\ 0.500,\ 0.666,\ 0.833,\ 1.000\}$$

Poisson Regression

$$\log(\mu(x)) = \beta_0 + \beta^T x$$

$$\ell(\beta|X, Y) = \sum_{i=1}^{N} \left[y_i(\beta_0 + \beta^T x_i) - e^{\beta_0 + \beta^T x_i} \right]$$

Penalized Objective

$$\hat{\beta} = \arg\min_{\beta_0, \beta} \quad -\frac{1}{N}\ell(\beta|X, Y) + \lambda \left[\frac{1-\alpha}{2} \sum_{j=1}^{p} \beta_j^2 + \alpha \sum_{j=1}^{p} |\beta_j| \right]$$

- **Penalty (λ):** Total amount of regularization.
- **Mixture (α):** Proportion of L1 regularization.
  - $\alpha = 1$: Pure lasso
  - $\alpha = 0$: Ridge regression

glmnet Poisson regression documentation link
parsnip Poisson regression documentation link